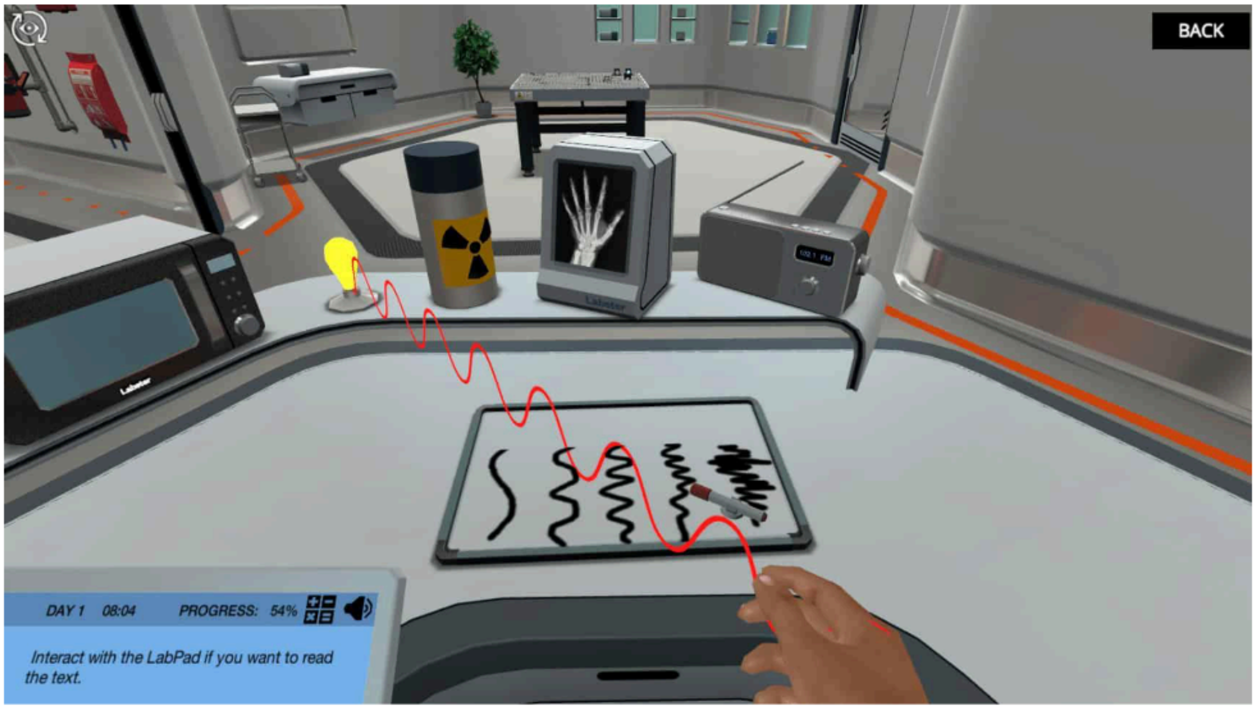
Electromagnetic Spectrum (Principles): Discover electromagnetic waves

In this assignment, you will be playing an immersive online learning simulation.

Below you can read the main learning outcomes and techniques you'll practice, as well as a brief introduction and storyline to the simulation.

Once you are ready, you can start the simulation by pressing the simulation launch button/link.


BACK
DAY 1 08:04 PROGRESS: 54%
Interact with the LabPad if you want to read the text.


BACK
DAY 1 08:05 PROGRESS: 87%
Identify the objects that emit electromagnetic waves that are not dangerous to humans.
Interact with the LabPad for more information.

Learning Objectives

At the end of this simulation, you will be able to…

- Use the electromagnetic spectrum to classify waves based on their wavelength and frequency
- Examine the relationships among the frequency, wavelength, and speed of waves
- Evaluate the damage to living cells by electromagnetic radiation

Techniques:

- Interactive animations

About this Learning Simulation

Electromagnetic radiation is all around us. In this simulation, you will learn about the electromagnetic spectrum. Join your virtual assistant Dr. One in a physics lab where you will interact with different types of electromagnetic radiation to find out their uses in everyday life and potential dangers to living things – don't worry, you will be perfectly safe in the virtual lab!

The electromagnetic spectrum

Join Dr. One in their physics lab and find out what links UV radiation to the microwave in your kitchen! You will learn how to define the waves that make up the electromagnetic spectrum and the two wave components by interacting with diagrams and quizzes.

Wavelength and frequency

See how everyday objects produce waves of various frequencies and wavelengths that define the electromagnetic spectrum. Discover how speed, wavelength and frequency are related. Some of the waves are visible, while others are not. But don’t worry, even though you cannot see them in real life, there are no restrictions in a virtual simulation!

Uses and dangers of electromagnetic radiation

Interact with different objects to find out how electromagnetic radiation is used outside of the lab. Food preparation, communication, cancer therapy, medical imaging and even just seeing the world around you all involve electromagnetic radiation. Some types of radiation can cause damage to living cells, so they will need to be handled with care – but don’t worry, Dr. One will keep you safe and teach you how to stay safe in the real world too!